Magnetic Moment in a Helmholtz Field

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A current loop in a uniform magnetic field experiences a torque $T = cI_H nI_L A \sin\alpha$, where cI_H is the field of a Helmholtz coil pair. This relation was tested by measuring the torque on circular loops with a torsion dynamometer while the coil current I_H, the number of turns n, the loop current I_L, the orientation α and the loop diameter d were varied in turn. The torque was linear in I_H, n and I_L and scaled as $d^{1.84\pm0.14}$, consistent with the area dependence d^2. The angular sweep followed $\sin(\alpha+\delta)$ with $\delta = (4.9 \pm 0.3)°$, an offset of the angle scale well beyond the $\pm1°$ setting uncertainty; a straight-line fit against $\sin\alpha$ would have underestimated c by 8%. The five sweeps gave $c = (7.03 \pm 0.11) \times 10^{-4}$ T/A, within 1.6% of the value $(6.92 \pm 0.17) \times 10^{-4}$ T/A predicted from the coil geometry. With this constant, a star-shaped loop of one turn has an effective area of (106 ± 3) cm². This exceeds the 70 cm² enclosed by its outline, as expected for a wire that crosses itself and encircles the central pentagon twice, but also exceeds by 16% the 91 cm² estimated for a regular star from its one recorded dimension, so a full measurement of the star's shape is needed to decide whether the two agree.

1. INTRODUCTION

A conductor carrying a current in a magnetic field experiences a force, and a closed current loop experiences a torque that turns it towards alignment with the field. This torque drives electric motors, moving-coil meters, loudspeakers and relays [1]. Because the torque depends on the loop's current, number of turns, area and orientation, and on the field, a measurement of the torque under controlled conditions tests each of these dependences, and a loop of known geometry can serve to calibrate the field.

This experiment measured the torque on circular loops placed in the field of a Helmholtz coil pair while each parameter was varied in turn. The sweeps test the predicted proportionalities and give the Helmholtz constant c, the field per unit coil current, which was compared with the value expected from the coil geometry. The calibrated field was then used to determine the effective area of a star-shaped loop of unknown area, as the manual requires [1].

2. THEORY

A magnetic moment $\vec{m}$ in a uniform field $\vec{B}$ experiences the torque [1,2]

$$\vec{T} = \vec{m} \times \vec{B}, \quad T = mB\sin\alpha, \tag{1}$$

where α is the angle between $\vec{m}$ and $\vec{B}$ (Fig. 1). A planar loop of n turns, each enclosing the area A and carrying the current I_L, has the moment $m = nI_L A$ (Fig. 2a), and a circular loop of diameter d has $A = \pi d^2/4$ [1]. For a loop that is not planar, or whose wire crosses itself, the area is replaced by the vector area [1]

$$\vec{A} = \frac{1}{2}\oint \vec{r} \times d\vec{l}, \tag{2}$$

taken along the wire. Each region of the plane then counts as many times as the wire encircles it.

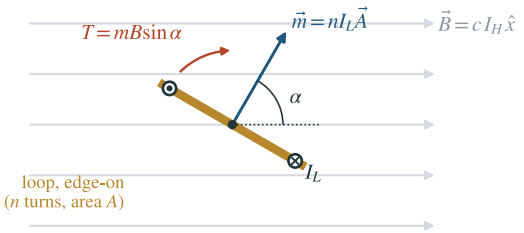


FIG. 1. Torque on a current loop, viewed along the vertical suspension axis (dot). The loop, seen edge-on, carries a current I_L out of and into the page at its two ends. Its magnetic moment $\vec{m}$ is normal to its plane and makes the angle α with the horizontal field $\vec{B}$; the torque $\vec{m} \times \vec{B}$ turns $\vec{m}$ towards $\vec{B}$.

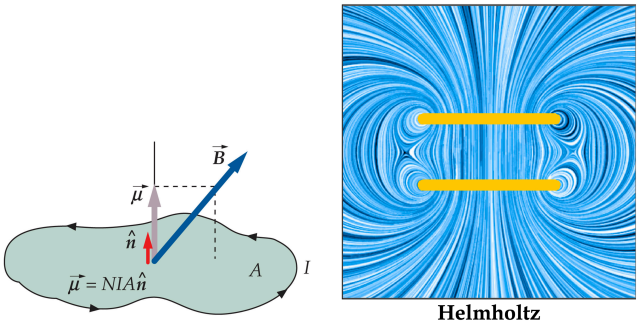
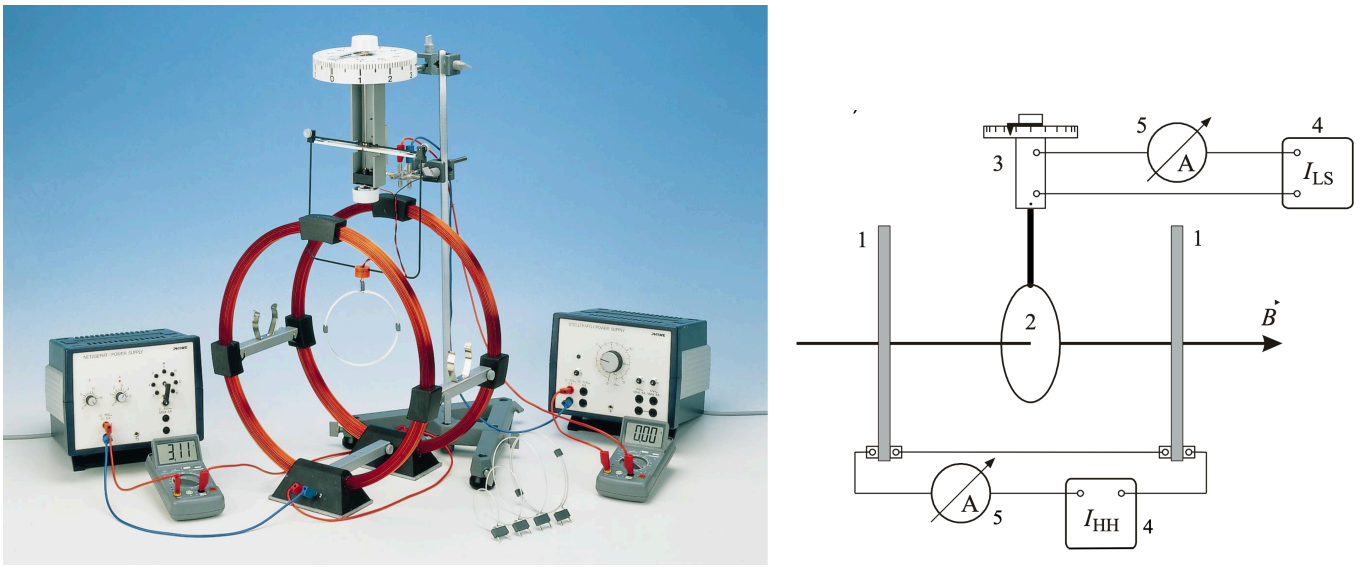


FIG. 2. (a) A planar loop of area A carrying the current I has the moment $\vec{\mu} = NIA\hat{n}$ along its normal; in a field $\vec{B}$ it experiences the torque $\vec{\mu} \times \vec{B}$. (b) Field lines of a Helmholtz pair (coils in yellow, seen edge-on): between the coils the lines are nearly straight and evenly spaced, so the field there is close to uniform.



(a) (b)

FIG. 3. Apparatus. (a) The Helmholtz coils with the loop holder at their centre, the torsion dynamometer above it, and the two power supplies and multimeters. (b) Circuit: (1) Helmholtz coils carrying I_{HH}, (2) conductor loop carrying I_{LS}, (3) torsion dynamometer, (4) power supplies, (5) ammeters. After PHYWE, experiment P2430400 [3].

Two coaxial coils of radius R and N turns each, separated by R and carrying the same current I_H, produce near their midpoint the nearly uniform field of Fig. 2b [1]

$$B = \frac{8\mu_0 N I_H}{5\sqrt{5}R} = cI_H, \tag{3}$$

which is uniform to fourth order in the distance from the centre. The Helmholtz constant c is in T/A for R in m and $\mu_0 = 4\pi \times 10^{-7}$ T m/A. Combining these relations gives [1]

$$T = cI_H n I_L A \sin\alpha. \tag{4}$$

The torque is therefore proportional to I_H, n, I_L and $\sin\alpha$ with the other quantities fixed, and to d^2 for circular loops. The slope of each sweep, divided by the fixed factors, gives c.

3. METHODS

3.1 Apparatus

Fig. 3 shows the setup. The Helmholtz pair had $N = 154$ turns per coil [1] and a radius of $R = 20$ cm (coil diameter measured as (40 ± 1) cm). The coils were connected in series, and I_H was kept below the 3 A continuous rating [1]. The circular loops from the loop set (one, two and three turns of diameter 11.98 cm, and single turns of 6.00 cm and 8.47 cm) and the star-shaped loop were mounted in a rotatable holder with notches every 15° [1], suspended from a torsion dynamometer (rated 0.01 N) at the centre of the pair. Two power supplies and two digital multimeters set and read I_H and I_L to 0.01 A. The leads to the loop were twisted together and left hanging loosely, so that the forces on them would produce no net torque [1].

Before each set of readings the dynamometer was balanced with both currents off: the dial was set to zero with the top knob, and the lever was brought between the marks on the fixed plate with the bottom knob [1]. With the currents on, the top knob was turned until the lever returned between the marks, and the dial reading F was recorded. The torque follows from the manual's coefficient $K = 11$ cm as [1]

$$T = KF, \tag{5}$$

so a reading in mN gives the torque in mN cm (1 mN cm = 10^{-5} N m). Readings were recorded to 0.05 mN.

3.2 Procedure

TABLE I. Settings of the six measurements. Unless varied, $d = 11.98$ cm and the angle mark was at $\alpha = 90°$ ($\pm1°$; $\pm2°$ for the star loop). Three dial readings were taken at every setting.

Sweep	Varied	n	Fixed
I_H	I_H = 0–2.82 A (7 values)	3	$I_L = 5$ A
n	$n = 1, 2, 3$	–	$I_H = 2.5$ A, $I_L = 5$ A
α	α = 0–90°, steps of 15°	3	$I_H = 2.5$ A, $I_L = 5$ A
d	$d = 6.00, 8.47, 11.98$ cm	1	$I_H = 2.5$ A, $I_L = 5$ A
I_L	I_L = 0–5 A, steps of 1 A	3	$I_H = 2.5$ A
star	–	1	$I_H = 2.5$ A, $I_L = 5$ A

Table I lists the settings. In each sweep one quantity was varied while the others were held fixed, and three dial readings were taken at every setting. The diameter of each circular loop was read three times; the readings for the reference loop were 11.95, 12.00 and 12.00 cm, giving $d = (11.98 \pm 0.02)$ cm and $A = (112.8 \pm 0.3)$ cm². The three-turn loop was used for the I_H, α and I_L sweeps because it gives the largest torque [1]. The star loop, one turn of wire bent into a five-pointed star, was measured once at $\alpha = 90°$; the distance between alternate points of the star, 15 cm, is its only recorded dimension.

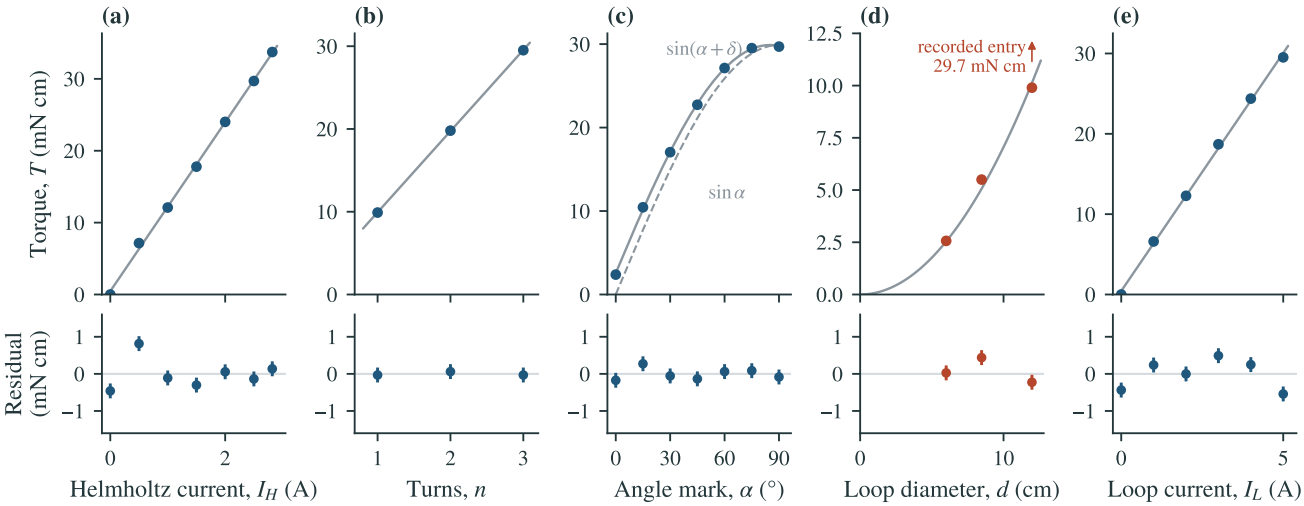


FIG. 4. Torque on the loop in the five sweeps (top) and the residuals from the fits (bottom). Each point is the mean of three readings, with the pooled uncertainty $\sigma_T = 0.20$ mN cm. Grey lines are weighted fits: straight lines in (a), (b) and (e); $T_{\max}\sin(\alpha+\delta)$ in (c), where the dashed curve is $T_{\max}\sin\alpha$ for comparison; and $T = kd^2$ in (d). In (d), the 11.98 cm point is the single-turn reading from the turns sweep; the entry recorded in the diameter sweep lies off scale (arrow).

3.3 Analysis

Each point is the mean of three readings converted with Eq. (5). Many triplets were identical at the 0.05 mN reading step, so the standard deviation of individual triplets is not a usable uncertainty. Instead, the standard deviation of a single reading was pooled over all 25 settings with non-zero torque, giving 0.032 mN (50 degrees of freedom). The uncertainty of a mean of three readings is then $\sigma_T = 0.20$ mN cm, assigned to every point.

The I_H, n and I_L sweeps were fitted by weighted least squares with a straight line, $T = sx + b$; the intercept b was left free to test for a zero offset. The angular sweep was fitted both with a straight line in $\sin\alpha$ and with $T = T_{\max}\sin(\alpha+\delta)$, where δ allows for an offset between the angle mark and the true angle. The diameter sweep was fitted with $T = kd^2$, the form Eq. (4) predicts, and, as a test of that form, with a power law $T = ad^p$. Parameter uncertainties come from the fit covariance, multiplied by $\sqrt{\chi^2_\nu}$ whenever the reduced chi-squared χ^2_ν exceeded 1. Each slope gives c by Eq. (4), for example $c = s/(nI_LA)$ for the I_H sweep; the angular sweep gives c from $T_{\max}$ and the diameter sweep from $k = cI_HnI_L\pi/4$.

The five values of c share the reference area, the dynamometer coefficient and, for four of them, the same point at $I_H = 2.5$ A, $I_L = 5$ A and $n = 3$. They are therefore not independent, and their fit uncertainties, which differ by a factor of seven, would not give fair weights. The unweighted mean was taken, with the standard deviation of the five values (not divided by $\sqrt{5}$) as its uncertainty. The uncertainty of the reference area and the 0.01 A current readings, which are common to all sweeps, were added in quadrature. The accuracy of K and of the multimeter calibrations is not stated and was not assessed.

The effect of field non-uniformity was estimated by integrating the Lorentz force over each circular loop in the Biot–Savart field of two thin coils of radius 20 cm, at $\alpha = 90°$, and comparing the torque with that in a uniform field equal to the central value. The star area follows from Eq. (4) with $n = 1$, $\sin\alpha = 1$ and the mean c. Its uncertainty combines σ_T with the uncertainty of c.

4. RESULTS

4.1 Dependence on Each Parameter

The torque was proportional to the coil current, the number of turns and the loop current (Fig. 4 a, b, e). The fitted intercepts, (0.5 ± 0.3), (0.1 ± 0.3) and (0.4 ± 0.3) mN cm, are each within two standard errors of zero. The turns sweep fits within the reading scatter ($\chi^2_\nu = 0.14$). The I_H and I_L sweeps have $\chi^2_\nu =$ 5.1 and 5.2: their residuals reach 0.8 mN cm, four times σ_T, without a systematic curvature. The point-to-point scatter therefore exceeds the repeatability of the three readings at one setting, and the slope uncertainties were scaled accordingly.

The angular dependence follows a shifted sine (Fig. 4 c). A straight line in $\sin\alpha$ fits poorly ($\chi^2_\nu = 6.9$) and needs an intercept of (3.0 ± 0.4) mN cm. The model $T_{\max}\sin(\alpha+\delta)$ fits within the scatter ($\chi^2_\nu = 0.71$) with $T_{\max} = (29.9 \pm 0.1)$ mN cm and $\delta = (4.9 \pm 0.3)°$. The torque at the 0° mark, 2.4 mN cm or 8% of $T_{\max}$, corresponds on its own to $\delta = 4.6°$, consistent with the fit.

The diameter sweep contains one inconsistent entry. The single-turn 11.98 cm loop was recorded as 2.70 mN in the diameter sweep, three times the 0.90 mN measured for the same loop at the same settings in the turns sweep and equal to the three-turn reading. With this entry, $T = kd^2$ fails ($\chi^2_\nu \approx$ 1000). It was therefore treated as a recording error and replaced by the turns-sweep reading. With the replacement, $T = kd^2$ gives $k = (0.071 \pm 0.002)$ mN cm/cm² ($\chi^2_\nu = 3.0$), and the free power law gives the exponent $p = 1.84 \pm 0.14$, consistent with 2 (Fig. 4 d).

4.2 Helmholtz Constant

TABLE II. Helmholtz constant from each sweep, with the fit uncertainty, and the combined value. A straight-line fit in $\sin\alpha$ instead of the shifted sine would give $c = (6.50 \pm 0.14) \times 10^{-4}$ T/A. The coil-formula value is Eq. (3) with $N = 154$.

Sweep	Fitted parameter	χ^2_ν	c (10^{-4} T/A)
I_H	$s = (11.75 \pm 0.18)$ mN cm/A	5.1	6.95 ± 0.11
n	$s = (9.81 \pm 0.14)$ mN cm	0.14	6.96 ± 0.10
α	$T_{\max} = (29.89 \pm 0.12)$ mN cm	0.71	7.07 ± 0.03
d	$k = (0.071 \pm 0.002)$ mN cm/cm²	3.0	7.19 ± 0.22
I_L	$s = (5.92 \pm 0.11)$ mN cm/A	5.2	7.00 ± 0.13
Mean			7.03 ± 0.11
Coil formula	$R = (20.0 \pm 0.5)$ cm		6.92 ± 0.17

Table II lists the values of c. The five sweeps agree to within a standard deviation of 1.4% and a full range of 3.4%. Their mean is

$$c = (7.03 \pm 0.11) \times 10^{-4}\ \text{T/A},$$

where the spread of the sweeps contributes 0.10×10^{-4} T/A and the reference area and current readings the rest. With $N = 154$ and $R = (20.0 \pm 0.5)$ cm, Eq. (3) predicts $(6.92 \pm 0.17) \times 10^{-4}$ T/A. The measured value is 1.6% higher, a difference of 0.5 combined standard uncertainties. Inverting Eq. (3), the measured c corresponds to a coil radius of 19.7 cm.

The Biot–Savart calculation shows that field non-uniformity is negligible at this level. For the 11.98 cm loop in coils of radius 20 cm, the axial field at the loop's rim is 0.8% below the central value, but the torque is only 0.2% below the uniform-field value; for the smaller loops the reduction is below 0.05%.

4.3 Area of the Star-Shaped Loop

All three readings on the star loop were 0.85 mN, so $T = (9.35 \pm 0.20)$ mN cm. With $n = 1$ and the mean c, Eq. (4) gives

$$A_{\text{star}} = \frac{T}{c I_H I_L} = (106 \pm 3)\text{cm}^2,$$

the area of a circle of diameter (11.64 ± 0.15) cm. A route that does not use c compares the star with the reference loop at the same settings: the torque per turn from the turns sweep, 9.81 mN cm for 112.8 cm², gives (107.5 ± 2.8) cm², in agreement.

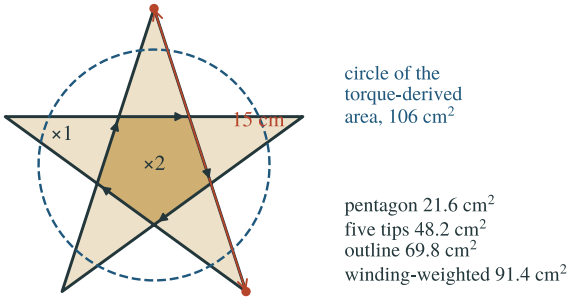


FIG. 5. Geometric estimate of the star loop's area, assuming a regular star with 15 cm between alternate points. The wire runs straight from each point to the next-but-one (arrows), so it encircles each tip once (light) and the central pentagon twice (dark). By Eq. (2) the effective area is the tips plus twice the pentagon. The dashed circle has the area obtained from the torque.

The wire of the star runs straight from each point to the next-but-one and crosses itself at the five inner vertices (Fig. 5). By Eq. (2), the central pentagon therefore counts twice. For a regular star with 15 cm between alternate points, each tip-to-crossing segment is 5.73 cm and each pentagon side 3.54 cm. The pentagon then has an area of 21.6 cm² and the five tips together 48.2 cm². The outline encloses 69.8 cm², and the effective area, with the pentagon counted twice, is 91.4 cm².

5. DISCUSSION

The torque-derived area of the star, (106 ± 3) cm², is 52% larger than the outline area of 69.8 cm². No plausible error in the torque or in c accounts for that difference, so the doubly encircled pentagon must be counted as Eq. (2) requires. The area also exceeds the regular-star estimate of 91.4 cm² by 15 cm² (16%), five times the uncertainty from the torque and c. This comparison, however, rests on a single length and on the assumption that the hand-bent star is regular. The photograph of the loop in the manual shows tips of unequal length [1], and the uncertainty of the 15 cm span was not recorded. Because the area scales as the square of the span, agreement would require a mean span of 16.2 cm, or an irregular shape with the same effect. The data therefore establish that the wire encircles the pentagon twice, but cannot decide whether the remaining 16% reflects the shape of the star or a further effect. The field non-uniformity over the star, whose points lie 7.9 cm from the centre, reduces the torque by less than 0.6% according to the Biot–Savart calculation, so it does not explain the difference.

The offset of $(4.9 \pm 0.3)°$ found in the angular sweep is five times the $\pm 1°$ assigned to the angle setting. It is consistent with the zero of the angle notches being displaced from the direction in which the loop's moment is parallel to the field, for example by the orientation of the holder on its rod. A constant torque offset, such as a force on the leads, would instead add a constant to every reading; the straight-line fit in $\sin\alpha$, which allows exactly such a constant, describes the data far worse than the shifted sine. The offset matters for the calibration. Ignoring it and fitting a straight line in $\sin\alpha$ gives $c = 6.50 \times 10^{-4}$ T/A, 8% below the shifted-sine value and outside the range of the other sweeps. If the 90° mark used in the other four sweeps was also offset by δ, those torques are low by $1 - \cos\delta = 0.4\%$, which is small compared with the 1.4% spread of c.

The scatter of the I_H and I_L sweeps, $\chi^2_\nu \approx 5$, shows that the repeatability of three readings at one setting underestimates the uncertainty of a point. Readings at different settings involve a new balance of the dynamometer, and a swinging loop or a drifting zero would differ between them. Since the cause was not isolated, the fit uncertainties were scaled rather than attributed to a specific source. The 1.4% spread of c between sweeps is consistent with this scatter. The remaining systematic uncertainties are common to all sweeps and do not appear in that spread. They include the accuracy of the dynamometer coefficient K and of the two multimeters, and the coil radius in the comparison with Eq. (3), whose 2.5% uncertainty dominates

the predicted value. The agreement with the coil formula is therefore a test at the 2–3% level.

The diameter sweep is the weakest test of Eq. (4). It has three points, the smallest torque (0.23 mN) is less than five reading steps, and one of its entries had to be replaced from another sweep. Its exponent, 1.84 ± 0.14, is consistent with the d^2 dependence but would not exclude a deviation of 10%.

6. CONCLUSION

The torque on a current loop in the Helmholtz field was proportional to the coil current, the loop current and the number of turns, followed $\sin(\alpha + \delta)$ with an angle-scale offset $\delta = (4.9 \pm 0.3)°$, and scaled with loop diameter as $d^{1.84\pm0.14}$, consistent with the area dependence d^2. The five sweeps gave the Helmholtz constant $c = (7.03 \pm 0.11) \times 10^{-4}$ T/A, in agreement with the coil formula, $(6.92 \pm 0.17) \times 10^{-4}$ T/A. The star-shaped loop has an effective area of (106 ± 3) cm²; the excess over its 70 cm² outline shows that the doubly encircled pentagon contributes twice, while the remaining 16% excess over a regular-star estimate could not be resolved from one recorded dimension.

Four changes would address these limitations. Tracing the star on paper and computing Eq. (2) from the coordinates of all ten vertices would replace the regular-star assumption and decide whether the torque and geometric areas agree. Locating the true zero of the angle by finding the orientation of zero torque, and setting α relative to it, would remove the 5° offset. Repeating each setting with a fresh balance of the dynamometer, rather than taking three readings at one balance, would measure the scatter that the $\chi^2_\nu \approx 5$ fits reveal. Finally, measuring the central field with an axial Hall probe would test c without relying on the dynamometer coefficient and the 2.5% uncertainty of the coil radius.

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